Yingxue Wang

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Education

University of Illinois at Urbana-Champaign Expected Dec 2026
Master of Computer Science GPA: 3.94/4.0

Beijing University of Technology, Fan Gongxiu Honors College Aug/2021 - Jun/2025
Bachelor of Engineering in Computer Science and Technology GPA: 3.7/4.0

Related Coursework: Computer Vision; Applied Machine Learning; Advanced Topics in NLP; Data Structure and Algorithm.

Publications and Preprints

Sara Aghajanzadeh, **Yingxue Wang**, Ieva Bagdonaviciute, and David Forsyth. *VLMs Win a Systematic Evaluation of Underwater Image Reconstruction.* arXiv preprint, 2026. arXiv:2608.11425.

Gao, J., Zheng, B., **Wang, Y.**, Liu, C., Liu, Y., and Guan, J. *Design and Implementation of YOLO v5-Based Supermarket Automatic Pick-up Robot.* Computer Era, 2023.

Research Experience

Graduate Researcher May 2026 - Present
Underwater Image Restoration and Evaluation, University of Illinois Urbana-Champaign

- Contributed to cross-VLM generation–restoration experiments comparing restoration behavior across generative model families.
- Developed an Unreal Engine component for underwater-effect simulation and restoration experiments.
- Co-authored *VLMs Win a Systematic Evaluation of Underwater Image Reconstruction*, an arXiv preprint (2026).

Undergraduate Research Assistant (Thesis) Jun 2024 - May 2025
Beijing University of Technology — Motor Imagery EEG Decoding

- Applied neural networks to the electroencephalography (EEG) decoding for motor imagery using Python
- Constructed decoding model and enhanced task accuracy by 5% using Transformer and Sinc filter

Undergraduate Research Assistant, Beijing University of Technology Mar 2023 - Dec 2023

- Analyzed app store user reviews for shopping applications using Python
- Captured explicit and implicit user insights for effective feature request identification using a customized BERT model (App Store Elicitation - BERT)
- Achieved high accuracy, strong F1 score of 0.94 and 0.83 for 'Problem Reports'

Undergraduate Research Assistant, Beijing University of Technology Dec 2021 - Mar 2023

- Developed an automatic pickup robot for supermarkets using YOLOv5 for object recognition
- Improved bounding-box localization and object-detection accuracy
- Achieved strong performance with mean Average Precision of 0.844 and F1 score of 0.95 for frequent classes
- Research resulted in a technical article in Computer Era (2023); see Publications and Preprints.

Technical Skills

- **Programming Languages:** Python, C, C++, SQL, JavaScript, Java, Verilog
- **Framework and Tools:** Git, Visual Studio Code, PyTorch, Jupyter Notebook, MySQL, Spring Boot

Professional Experience

TRS Information Technology Co.,Ltd. Beijing, China

Intern, Data Science-Python Data Mining Public Interest Aug 2024

- Participated in 'B&B Price Prediction' and 'Credit Card Fraud Detection' projects using Python
- Strengthened skills in data mining, machine learning and AI

Project Highlights

Multi-Turn Jailbreaking Evaluation for LLMs Aug 2025 - Dec 2025

- Developed an adaptive multi-turn jailbreak evaluation framework for LLMs using scripted and automated attack strategies to model realistic attacker–model interactions
- Evaluated GPT-3.5 and GPT-4o on 12 high-risk behaviors, quantified attack success and turn-aware harm severity, and analyzed multi-turn risks against single-turn resampling baselines

Investigating Faithfulness in Adaptive Reasoning CS 562 Course Project

- Implemented a GSM8K arithmetic-intervention pipeline comparing correct-prefix and corrupted-prefix continuations to study reasoning faithfulness and error recovery.
- Integrated AdaptThink inference with configurable continuation budgets and random seeds, recording answer correctness, token use, and traces for manual review of explicit error repair.

Competition Awards

National First Prize in the China Computer Gaming Competition (Top 5%) 2023, China

Meritorious Winner in Mathematical Contest In Modeling (Top 7%) 2022, USA

National Third Prize in 26th National English Debating Competition 2024, China